

Active Inference BookStream #001.08 ~ "Governing Continuous Transformation"

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hello and welcome it is January 31st 2023 we're here in octave bookstream number 1.08 off to you blue so we will be discussing this book continuing our discussion of this book governing continuous transformation by Jean kesri we are the active inference Institute participatory online Institute that is communicating learning and practicing applied active inference you can reach us at all of our social media links this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we will be following good video etiquette for live streams um again if you would like to participate in this project or any activities that are ongoing in the active inference Institute please reach out to us there are plenty of ways to get connected and involved um okay so the goal of today's bookstream is to continue our discussion of this book um and we will just be going through it as usual um just as always this is not a review or a final word over this material just as an introduction to some of the ideas um okay so just quick introductions for those of you who don't know us I am blue Knight I've been involved in the active inference Institute for quite a while I'm an independent researcher in New Mexico and I'll pass it off Daniel hi Daniel I'm a researcher in California I'll pass to the title my name is Tyler I'm a critical designer and GAP researcher in California right So today we're moving on to chapter eight of the book uh the title of this chapter it's in part two the free energy governance section and the title of this chapter is capabilities Duality management and we'll talk more um in as we go on as as to what Duality management kind of means um okay so there are two sections outlined by Bijan here in this chapter there are Dynamic capability View and antagonistic neural networks perspective there's a whole bunch of keywords and we'll just jump right into it um Tyler do you want to read this part of the abstract for us sure all right so abstract Dynamic capability view GCV and antagonistic neural networks perspective A and P address free energy governance capabilities Dimension the board's cognitive capabilities in terms of meta managing interdependently opposing Logics as Unity are increasingly relevant

to continuous strategic renewal Duality management such as the vacillation between exploitation and adverse exploration and balancing financial profusion control is established as a dynamic board capability effectively FEG extends dev to the board level more importantly by adopting a free energy principle power perspective by redefining the firm as an inactive inference machine dcb's triplet of sensing seizing and reconfiguring S has this operationally such that the firm is engaged as a whole great so antagonistic neural network perspective is established and applied as the neuroscientific basis for Duality management managing the existing while contradicting the present to develop the future this task positive Network slash uh what is the other one gmn Dynamic mode Network I can't remember uh antagonism what is it default mode Network there it is um default mode Network antagonism between exploitative task execution and explorative explorative abductive thinking is a neurological representation of the contradictory demands that challenges companies sustainable outperformance and survival the board is traditional monitoring and advisory roles must be explicitly complemented with The Duality management mandate um managing both of these networks more specifically the board's Dynamic capability of Duality management lays the foundation for the board's newly established governance purpose and discontinuous and distributive distributed markets quote Duality stewardship um yeah and so we'll get into these task positive and default mode networks as we get in later to this presentation all right um so this is a dynamic capability view I looked it up I found this cool image um so Bijan says sensing strategic opportunities in Dynamic Market environments seizing the opportunity and reconfiguring a firm's resource base accordingly are a dynamic capabilities use three Central pillars and he cites this reference um so here we see these fencing seizing and managing or I guess that's reconfiguring um these are the three pillars and then they give some micro foundations like what is this sensing seizing reconfiguration or managing threats um so I don't know Tyler do you have any like thing to add here or maybe like specific examples um I mean I see like I I to me this is a very abstract like diagram and I'm like what's going on here yeah I mean I think this makes it more seem more difficult and complicated than it really is where a lot of times with sensing a lot that's where there's like that abductive reasoning we're saying like hey there's something here what could be true here like how do I explain kind of this like these disparate signals I'm getting from the environment how do I make sense of this and where seizing is like actually building the thing and managing Progressive performing is like managing that thing you built over time and I think you actually in the next slide it gives a really good example with um Amazon Kindle I've actually maybe if you could just go to the

next slide I think it um explains it well so for example it says for example Amazon's launch of the Kindle 2006 involves sensing and the impending threat of its Core bookselling Business it's using the opportunity through Rapid development of the first generation eBook reader and gradually reconfiguring its internal activities to ensure that the Kindle became an integrated part of its overall Market offering thanks thank you um uh bishan goes on to say in a similar manner to that in which organizational Learning Resource research distinguishes between single Loop and double Loop learning capability research also distinguishes between operational capabilities and dynamic capabilities and so um in the text he gives some examples of like operational capabilities like mergers and Acquisitions and other like things that are outside the scope of this Dynamic capability View um okay so start reading these claims so he says there has lately been a shift in Dynamic capability view from behavioral to cognitive interpretations explicating the cognitive micro foundation's underpinning thesis 2007 Three core pillars of dynamic capabilities sensing perception seizing sense making and asset configuration language communication and social cognition so this is like reframing again those those three pillars kind of more towards the active inference of free energy governance perspective um he was on to say with respect to Dynamic managerial capabilities for seizing and reconfiguring in particular we have emphasized ways in which top management can shape aspects of the organizational context such as by encouraging cooperation lowering resistance to change designing a business model for a new Venture and investing in new skills and assets this suggests that managerial cognitive capabilities May function as mediators of the relationship between changes in organizational context and strategic change which in turn can affect firm performance do either of you guys have any comments here should we move to the next one okay all right so I had to look this up because the term industrial vertical was like brought up and I was like what is industrial vertical um and I think like it's more commonly called an industry vertical you could have it like in the industrial field and then it's like something different but an industry vertical is also known as a vertical market and it's like a unique business Niche that serves a specific set of consumers or customers and their particular needs and there's like tons of examples of that like I don't know organic food and I don't know like solutions for a biology lab or like there's you know all these like Niche markets and and it's like if you're trying to sell chemicals to a biology lab like you know you're selling them like hplc purified acetic acid and like you're not selling that acetic acid to you know someone to use in their kitchen for salad dressing like it's a totally different like even though you're selling acetic acid those are two totally different markets of people so you don't want to make

a product that's like bought by everyone because then it actually is not useful for anyone that was kind of my understanding do you have anything to add Tyler yeah it makes it could give like a little more color of examples here just like an example of an industry vertical could be like movie making business or like on the top end you might have like where the movies are consumed it's a movie theater and then you have how it's produced so you have Studios you have screenwriters uh you're the people who are actually creating the equipment and making the film right so it's basically like this big kind of almost supply chain or these interrelated businesses that are all supporting each other to one um final product so I one thing I would add here too that I think the Jean would probably agree with is that in a more discontinuous world the idea of an industry vertical is becoming a little bit more fuzzy and so for example like in the past you might say like well I have this I make research I make chemicals for biology research right but it could be the case that you also make chemicals that are used for like hair products and so like your verticals are increasingly becoming more fuzzy and intertwined and it becomes harder to think about like like where do I actually like what Niche do I actually serve like what is the market I'm in right and it could be really hard to think that and that's where sensing really comes in handy is because you could be like oh well I thought I was making biology research chemicals but it turns out because I've I'm sensing and using my adductive reasoning and when they've you know my best fpg perspective I can I'm actually in the hair care market and that's where I really should be um and so it's kind of the idea of a really cool becomes harder to Define uh over time this website actually gave a lot of examples about how uh like industry verticals allow companies that would sometimes be competitors to um collaborate with each other like they talked about how the Samsung glass is used by the Apple iPhone or maybe it's the other way around but either way like Samsung and like the Samsung phones and the Apple phones are normally like you know competitors in a market um I don't even know that they compete anymore like either you're a Mac person or you're an Android person like I don't think that there's any competition there but maybe so but how the you know they're collaborating with the use of the glass and making it accessible to everyone just because everybody needs glass right um okay so they also Bijan also brought up um oh yeah here is the industrial vertical so let me read this from the from the book it says indeed independently of industrial vertical the increasingly distributed and discontinuous nature of Market transforming forces cannot be overestimated however Market velocity velocity is too generic of a concept and so like I was trying to understand like how he was using the term Market velocity like here and like so Market velocity like when you Google it it's like it

indicates the level of Market activity in individual NASDAQ stocks by measuring the frequency and share volume in aggressively priced orders it can indicate changes in Market Direction momentum or liquidity in each stock enabling users to imply future price movements And Trades so like that's how like Fast the Market's moving I guess um and then like it's also a company that recycles and like velocity is a measure of two Dimensions right the speed of change and the change in Direction so I don't know um like the concept of Market velocity here like he can he goes on to compare it to environmental scarcity so the more suitable um Dynamic capability concept is environmental scarcity uh inspired by Cameron and zamuto 1983 meaningfully relating sensing and sense making as meta capabilities respectively to econation Dynamics and strategic renewal so I don't know um like how environmental scarcity and Market velocity like how those two relate to each other I don't know if you guys can offer any insight here um but this is something that like I would hold out and ask John about yeah I think with with Market velocity like a lot of times when people talk about discontinuities and like like people who are not coming from an fpg background I'm talking about hey like things are moving really quickly it's more uncertain than ever they talk about Market velocity right like things are just changing all the time and I think what vagina is saying is like just saying that things are changing very quickly isn't very helpful to you because you're like okay cool like life is hard I get it like now what do I do with that information where when you presents environmental scarcity it's becoming a little bit more specific and like how you can actually be more action oriented with it because you're like okay well like what's my Niche and what are the meta capabilities I have to to create to manage this uncertainty and create my own Niche uh whereas Market velocity is kind of like being overwhelmed by uncertainty and being like I don't know things are hard like this is scary that's at least my take so she goes on to um oh sorry go ahead one related angle to connect back to what we've discussed elsewhere is this concept of generalized coordinates which is the value that something is at and then the first derivative which is the velocity and then the acceleration higher derivatives and so on and so that allows you to track one variable like the temperature or the salinity of the water or the funds in the bank account and look at it not just where it is but how fast it's moving and how fast it's changing how fast it's moving and then you can learn certain things like oh the third derivative is always within this range so that puts certain bounds on timelines that the primary indicator might change by a certain amount or not and we see some of those things like um the gradual decline sudden decline those are differentiated not by their starting point but by the rate of their decline so velocity is something that we can

talk about in Bayesian physics actually as a speed um and then he goes on to outline this environmental scarcity um from this Schmidt at AI 2016 reference which says combining the type and rate of decline allows us to distinguish four environmental scarcity situations environmental dissolution which is the gradual decline in shape environmental erosion gradual decline in size environmental collapse sudden decline in shape and environmental contraction sudden decline in size so like I was trying to like picture this as like a like a um I don't know maybe like a market right like and so like the market changes shape like what would it mean for a market to change shape really um I get like the market changes size like all of a sudden you know there's not a market for CDs anymore because everybody's listening to mp3s so like that's like a well I guess like that's a change in shape and size I'm not sure how to really think so so to use that example like that would be a change in shape where like for example imagine the same and number of people were listening to music but the format changed so it's like the the market is music people who listen to music but the shape is a little bit different of like how they're consuming it whereas it could be a try to change its size and shape as if like you move to mp3s now because it's MP3s more people can access the music the market grows and that'd be a change of size and shape or like the market totally disappears because we can just stream whatever we want on YouTube now like yeah exactly that's like that's also a change in size and shape I guess um but yeah okay cool all right so uh environmental scarcity is more suitable than Market velocity and I guess I can see why that that is because it it's more nuanced as to like what is actually changing is it just like up or down and how fast or like the shape and the size um and like the rate I guess all of that is is taken into this environmental scarcity concept um okay uh do you want to read these claims Tyler sure so while slow and gradual Market changes tests in particular our sensing capability more certain changes challenging firms sense making capability uh there's really is no objectively uncertain Market environment uncertainty is an ad for dependent mental experience facing one's very limitations of dynamic AirPlay between sensing and sensor making therefore independent of the degree of Market philosophy Dynamic capabilities are critical he goes on to say the stronger and setting the impact of environmental scarcity the higher the urgency to act and hence the stronger the tendency for strategic renewal to be rather Financial control center I M A and other forms of alliancing precisely for two reasons one these the existing resource base is too trapped in the past to unlearn Advanced renewal from within and without or to the opportunity for Market consolidation is too irresistible and as JP Morgan Had demonstrated during the 2007-2008 financial crisis and time to action is critical cool um so yeah I guess that that kind of sheds

more light on the environmental scarcity um concept here uh okay so just I wanted to Define this we kind of talked about the tyranny of the ore um last time but we didn't really elaborate on it and so we we last time we talked about this good to Great to Gone book um Circuit City and uh the tyranny of the ore is actually um Bijan keeps quoting this wurtzel book about the tyranny of the Lord but I had to like look it up it's actually from the not good to Great Jim Collins book but like another Jim Collins book that I haven't read uh built to last and so here the tyranny of the ore is the rational view that cannot easily accept Paradox that cannot live with two seemingly contradictory forces or ideas at the same time the tyranny of the ore pushes people to believe that things must be either A or B but not both um and so I think that this concept kind of goes like is like a light introduction and we'll dive deeper into it about like Duality and Duality management like being able to hold two simultaneously um opposing or maybe seemingly opposing or necessarily contradictory opinions and reviews ah okay and here we go right into uh Duality management um so Vision says in fact at any point in time a firm is challenged to balance strategic organic r d driven Innovation and growth with financial control external growth and Innovation by way of forms of strategic partnering there is no tyranny of War it is The Duality a continuous balancing act that is mutually reinforcing and with perspective and time and space forms a Unity free energy governance establishes Duality management as a dynamic capability specifically as a dynamic board capability Duality management such as vacillation of exploitation versus exploration and balancing financial and strategic control is established as a dynamic board capability the goal-directed action-centric and generative circular causality of top-down bottom-up and lateral sense making processes prediction error minimization forms the basis for leveraging opposing Logics in the form of duality um and Duality Management in contrast with uh strategy management do you want to read this one Tyler sure so indeed strategy management has traditionally embraced one environmental co-creation versus co-alignment two exploitation of the first exploration and three Financial for strategic control as dualism or Paradox rather than as Duality it's the background of a world that is increasingly distributed just continuous and in this capability interdependent persistence persistency and contradiction of interdependent elements which is the defining core Paradox it's potentially nonsense and therefore severely limits the theoretical as well as practical relevance of a strategic Paradox lens foreign you know the old way and as opposed to Duality management which is um the new way do you guys have anything to add here this is kind of like Paradox is attention to balance Unity is plural and at minimum two rather than Paradox or something that must be resolved one way or the other so it's really

compatible with a lot of qualitative things people say about wisdom and balance and I think the question will be how is that enacted in organizations and how does that connect to active yeah I mean one thing that's interesting about Duality management is that there's also like a strong cultural tendency to be decisive or a biased to be decisive um in that like the perception of kind of going not wanting to make a choice can be seen as weak um whereas like you haven't really so like there's even if I think a lot of people would recognize The Duality and say like yeah like I don't know it's kind of both like you want to explore or you want to exploit like I think a lot of people would would recognize that but when you're in a meeting and you're not like really committing to something well you're not like really a lot of people feel like they're not really offering Like A New Perspective like they're not putting their stake in the ground saying like this is where I how I feel about it and this is where I'm gonna like you know push behind uh and so I feel like a lot of people feel this pull to like make a call and so that's one like even at an executive level you can feel that especially two-year like on the executive level and even if you recognize this Duality and you're communicating what you want to do to the broader organization like Duality is a really hard thing to communicate and say like yeah we don't really know what's going on like it could be one of these multiple things like let's figure it out together I have done this look I have like communicate this before it's A Hard Road to go down right we're like hey yeah we're all just figuring it out like it's gonna be like let's just kind of be fluid about it a lot of people don't like to hear that because it sounds like well this person doesn't know what they're doing and I don't even like know where we're going this is really scary um and so I do want to touch on this later but like it's one thing recognizing duality in yourself and on a small management team how you talk about duality in a broader organization there's like a whole different animal um and so I hope we can talk about that more a little bit later yeah thanks for that uh okay so um we talked last time also about um like the quantum Duality and like what what does patreon mean and like we knew that he would really kind of um unwrap it here in this chapter so um I I guess like we've seen Duality like we um are able to like hold simultaneously it's simultaneously a one and a zero and so we were wondering if if uh what was really meant by Duality like in the quantum sense was more like the superposition um angle and so it it seems like yes um uh based on even this quote from the book sober Jean says it is important to appreciate that many dichotomies that have Divine defined the science and practice of strategy management such as firm versus environment executive leadership versus non-executive Board individual versus firm exploitation versus exploration Financial control versus strategic control formal versus

informal channels and flexibility versus stability for example are less and less dichotomous in a distributed and discontinuous world but coexist in superposition as Duality or synchronicity so I think that this kind of maybe answers at that question for us from last week if you guys have any comments here okay so oh Daniel yeah some of those dualities are raised as a partition like firm from environment or organism versus environment it's a partition that is an integrated whole and so that is part of this framing question seeing it as an integrated unit but also containing um separate aspects and then also Quantum carries a lot of baggage whether somebody's familiar with what it technically means or not dualities in the quantum space could refer to wave particle it could refer to Observer observed and so I think it needs a lot more operationalism to have utility Beyond a sort of like evocative Quantum management and then again to come back to Tyler's point which I want to hear more about if you say well we're actually doing all the strategies it's like a wave you thought we were going to go to one strategy or the other we're going to be probabilistic across all of them how does that work Tyler I'm waiting for your comeback oh well I mean honestly like I have actually strategy of how I do this and I just don't necessarily communicate broadly that we're doing oh the strategy is probabilistically like like I might know that and maybe a couple other people know that we're like kind of like keeping our options open for a bunch of different Avenues but you communicate something that's much more clear um you can't I don't think you get to do both but you don't get to communicate that broadly it's just not consumable by other people and it doesn't give them like it's not really actionable by them so you can't say we're doing all the strategy there because they're like cool then how do I fit into this you have to say like well this is broadly what we're doing and this is how you're going to fit into it um and so yeah also action helps cut this knot so for any given search space it's not possible to have one parameter value that is maximizing exploitation and exploration that gets into the sort of like I'm everywhere on the number line at once type thinking but action allows us to say well our strategy is we're going to move quick but as soon as criteria one two and three are met then we're going to lock in so that's actually a plan that you can have Precision about that embodies in a another variable space explore exploit but where we have Precision about is about action which is what his whole emphasis is on actually and that reminds me another strategy I use too is that you explicitly frame it around exploration and that like you're waiting for something to exploit so you're saying like hey all right we're going to build this thing and we're not sure like who's gonna even use this right but we're able to say we have a hypothesis that these kinds of people might use it and then we're gonna see who uses it and that's like the experiment we're

running together as a company is this like Define scope explore stage and then once we run that experiment then we move to exploit and so you can like kind of like clearly do that within an organization of like how you're switching between these different modes nice okay uh Bichon says therefore to effectively operationalize Dynamic capabilities triplet of sensing seizing and reconfiguring we need a model that dynamically links Dynamic capabilities triplet with the environment this is free energy governance is underlying and active inference one active States acting sensing two sensory States perception sensing sense making and seizing the opportunity and three internal States firm resources resource configuration and four external States unknown hidden are all Linked In circular causality conditioned by the internal external Markov blanket and governed by free energy minimization and so I just thought it was interesting here like we can go back and look here at the three pillars like um the sensing seizing and like reconfiguring and like for me like the sensing part um like goes here like in this action sensing or sensing is like in sensory States but like seizing the opportunity as a sensory State like totally threw me off um like is it just me like a sensory like sensing the opportunity is like one thing but like leveraging an opportunity as like a sensory State like that seems more like an action State I don't know if it's just me or like what you guys think about that um because I was kind of like whoa seizing the opportunity is a sensory state well for example in seizing he included like creating your business model and so that is I think part of sense making because you're like this is how we relate to the ecosystem so it's like action like this action centered because there's also sensory or something like well this is the kind of like information that we're going to collect from the environment as well and so I think you're kind of doing both the action and perception when you're doing sense making here so like this is how I'm going to relate to my environment not sure if that's exactly was what you were saying I think this is an attempt to coerce for better and For Worse a triplet into the particular partition where blankets are defined as graph a Bayesian graph where there are nodes that make two other sets of nodes conditionally independent and I think we've seen in the philosophical and application settings a lot of challenges with people simply coercing a naturalistic understanding of how things are in the world with this Bayesian graph formulation like here we have active States action sensing action states are the outgoing statistical dependencies of an entity so yeah they are action and if you think about active sensing then they're sort of elusively related to sensing but it's not a sense State and so how much are we going to have both feet in the technical formalism of the particular partition and to what extent are we kind of leaving one foot on the formalism and then pivoting around and trying to draw in some other Frameworks and draw

concordances so I I agree that it's like a coerced um like like put moving the triplet into this like particular partition but I I just don't know if seizing the opportunity is something that I would include as a sensory State um I I think that that's something that I would put like even in constructing a business plan like that's still an action that you'd like Take On The World so so that's something that's outgoing not something that's necessarily incoming and that's just how I would frame it um or like the way that made more sense to me so I just found this a particular like a particularly curious um area to to put it um but yeah we can move on and talk about we can ask me John why he put it there uh hopefully and then we can talk about this antagonistic neural networks perspective um so Bhajan says antagonistic neural network perspectives is established and applied as the neuroscientific basis for Duality management managing the existing while contradicting the present to develop the future it is the form of organizational learning that Strategic Management literature has labeled ambidexterity co-managing exploitation and exploration in neuroscientific terms the antagonistic relationship is between two large-scale cortical networks the task positive Network tpn and the default mode Network dmN which simply put our neuroanatomical representations of top Management's day-to-day task specific focus on exploitation the task positive Network on the one side and the rather non-routine Divergent and creative thinking that initiates exploration and strategic renewal the default mode Network um and so I thought this was super interesting and took like a deeper dive into like looking up the reference for that uh quote um and so here in this paper antagonistic neural networks underlying differentiated leadership roles by boys and uh rashford and Jack they say the antagonistic relationship between the task positive Network and the default mode Network creates a fundamental neural constraint on cognition that is highly relevant relevant to the different roles and capabilities that effective leaders must astutely juggle and deploy an important consequence of this constraint is that an over emphasis on task-oriented leadership can prove deleterious to an organization in particular when openness to new ideas people emotions and ethical concerns are important to success on the other hand the over emphasis on relationship-oriented leadership may prove deleterious to focus and and the execution of clearly defined goals and I thought that this was really cool I just have never like thought about leadership like this but it's definitely there's like the people component and like the GSD component like getting stuff done um and and like leadership requires both and like you can't like you need to have both and I also thought like that the ambidexterity here like where Bishop talks about is like interesting because like you think about like organizations and you always hear like the phrase like the left hand doesn't know what the

right hand is doing and I wonder if like that's why this is like uh this different management um like role or dichotomy is like labeled like the ambidexterity or like what what which which came first the chicken or the egg here um I don't know to get through any comments here one thing I thought was interesting here is like you know one level there's like this analogy between like the default mode Network and the test positive network with we might find in an organization but I've always was interesting too was thinking about how the individual relates to the organization and that like you know the the creativity that an individual or leader can exercise is kind of constrained by the antagonistic relationship between their own test positive and default mode Network right and you know it's one thing for people to say like yes like you need to have this antagonistic relationship and it used to be this balance but the people that's the ability for the organization to do that is constrained by the people within that organization their own individual abilities to kind of switch between these two these two systems and you know it's it's kind of a lot yeah I guess you're basically like the organization's ability to do this runs up against the biology of like the individuals and their own ability to switch between these different modes and I think it's that ability to kind of run these two modes in parallel is quite rare and it's one thing to ask your leaders to have it but to ask your entire organization to have it it's like a whole much different and much bigger ask and so one of my things I keep thinking about when I'm going through this book is like to what extent does the full organization need to have internalized like Duality management or is it really just the leaders or I think what Bajan would say is like the board but I don't that doesn't seem to connect each level of the hierarchy like it seems like each level of the hierarchy should be able to manage these two different modes of processing but it's a rare skill so how is that even possible to do um yeah I'm not sure if you get any thoughts on that yeah a few more one is the switching which is sometimes even potentially like faster than a second of our own behavior or neurophysiology from task oriented to mind wandering and to what extent we sometimes just zone out thinking daydreaming reflecting integrating looking at our schedule looking at our inbox like it's not task driven but that is like a space that can then transition into a task not always the task we want but it can transition to a task so that's kind of cool and antagonistic's one way to frame it you could also call them synergistic or synergetic because they're part of that Unity is plural and at minimum to angle where no single neural network could be tuned to be able to be both diffusive and task oriented so why not have some type of oscillatory process that enables those two modes like having a forward in a reverse in a car instead of trying to make one gear that can do two things that are contradictory yeah it says super good point um

and in this paper they really uh talked about um they went in a greater detail about the task positive Network and the default mode Network and they said that the task positive network is activated during a broad range of non-social tasks and is thought to be important for problem solving focusing attention making decisions and control of action in other words for getting things done however activation of the task positive Network also has a deleterious effect on other cognitive functions that are essential to leadership it suppresses activity in the default mode Network and similarly the default mode network plays a role in emotional self-awareness social cognition and ethical decision making it is also strongly linked to creativity and insightful problem solving and then the activation of the default mode Network they said elsewhere inhibits the the task positive uh Network so I thought that there's like this like neat and it is like you know Unity you need both um you can't constantly like be focused on a task like you guys know how that feels right like to just be like uh I'm like out of it like you have to have that like you need to like have work and play um in order for some kind of balance or like it's it is unifying to be balanced I think just one thing that's made me remember I think I've actually mentioned the same previous stream but you know in regards to like switching between the two Gears of the car rather than trying to like run them both at the same time one way that this is was phrased to me by one of my previous uh CEOs was you have like art projects and you have science projects and so like your art being your default mode projects your science being your task positive projects and like explicitly saying like we are in art mode right now like don't hold us accountable to any metrics like this is an art project we're exploring uh versus science which is like hold me accountable I mean and that means hold me accountable to metrics like I'm going to set a number and I'm going to hit that number and like that's how you I should be evaluated and so I think there are ways you could probably like frame switching between these modes in an yeah thank you Tyler and one biological angle on this sometimes this type of oscillation is called a winnerless competition and it's modeled using Predator prey type Dynamics uh Lotka Volterra is the name of the dynamical formalism and in the SPM textbook and package from Carl Friston and colleagues earlier work there's a lot of examples of winnerless competition lot of Volterra models being used to model activity amongst different brain regions and this winnerless competition is an interesting framing on everything from attention to sleep it's like sleep is not trying to be a hundred percent of the time wakefulness is not trying to be 100 of the time they inhibit each other until there's a phase transition and then it's like a dwell in one state and that inhibition shifts and one oscillates back and so similarly like a predator doesn't over the long run really want to

eradicate the prey but their populations fluctuate fluctuate and that's a winnerless competition and in fact that's a co-winning competition and just one last Point um I remember in the selfish Gene some of the discussion of out there on the Savannah look at the cheetah eating the gazelle how can one say that that's like a mammal level selection at play that doesn't make any sense it's like but actually if that ecological relationship weren't there then neither population would have the resilience so how we frame things and especially the level of holism that we bring is going to set the stage for whether we see something as sustainable or unsustainable adversarial or non-adversarial awesome okay so moving on uh Bijan says free energy governance is firmly rooted in the belief that in a discontinuous and distributed world the marginal benefit of structural reorganizations as well as structural compartmentalization of exploitation and exploration are rather sub-optimal or possibly counterproductive competitiveness and survival are increasingly determined by the extent to which a system is rather less rule or top-down supervised learning based exploration happens increasingly in real time within exploitation in effect the emergence of the new is essentially hypothesis testing within the existing and I really liked this like um uh like Daniel you had mentioned earlier about like is this a partition or is this like a nesting um and like this exploration and exploitation um is like is a partition and also is like a nesting um based on on what Bijan says here uh he also says in a distributed and discontinuous Market environments the real leadership challenge is to de-antagonize the Past positive network default mode Network dichotomy in the form of Duality management um do you guys have any comments here moving on do you want to read these sure feg's cross hierarchical connectivity logically concludes that managing the tpn DNN antagonism it's not a top-down challenge calling for structural compartmentalization but one of Duality management by vacillating between or among discrete formal organization mode such as centralization and decentralization the organization may increase dynamically the levels of exploration and exploitation Beyond those achievable through an approach based strictly on a static design Choice summary pretty much conceived organizing as an informally deep learning and self-designing Compass centralizing intelligence at any hierarchical level of the organization can be counterproductive increasing free energy promoting entropy and eventually capitulating to tpn DMS and antagonistic forces um so I think one thing you might have been referencing here is when you're structurally trying to create you know your DNN and tpn and like having them separate from each other this is what's like traditionally done in companies of having like an r d department where like this is where all the like the new ideas come from um and so it's I think

he was juxtaposing the fpg approach to like that very traditional model oh so we'll move on here so he gives here the free energy governance response to the task positive network default mode Network Challenge and it's this three-fold response and he says sensitizing the entire organization in regard to Upper echelon's tendency to top down extrapolating is dominant supervising knowledge structures logic and past experience as a guide and prediction for future success a fallacy that tasks specific exploitation tends to reinforce embracing strategic renewal as a matter of continuous deep learning in the form of cultivating cross-hierarchical processes empowering purpose-directed and action-generated bottom-up Market stimuli as a basis for double Loop learning question the very models that generate the top-down predictions and three default abductive attentional engagement at every level of the organization forms the basis for Effective top-down bottom-up prediction error minimization after noticing an environmental change it is essential to question one's understanding of how the subsystems of the environment EG technology and markets and the organization EG knowledge and goals are linked without losing sight of the fact that environment is not objectively defined but a product of action generated and socially constructed shared meaning and expectations so I think this is like how the Duality management works essentially um that's kind of what what I took from here Tyler do you have comments yeah and just that like these are all really big asks to make up an organization at every level like I think it's one thing to say like yeah I agree that this is the answer and if this could be done like be like hell yeah like I'm all for it but I again feel like you're kind of running up against people's individual abilities to manage Duality just within themselves um and that will constrain your ability to do this with an organization because I think about the cognitive cone right like uh when we that we talked about with Michael Levin and um it's funny like what you just said is like the flip of the cognitive cone so usually it's like the you know the the larger structure or like the more top level hierarchical structure will cognitively constrain like things that are nested within it so the shape of that cone the the shape of the individual is within that cone can't um be any other shape right like so they can't come out of the cognitive cone of the entire organization but but it's interesting to think about how what is happening at the individual level like the shape of the individual cognitive cone also uh lends itself to the configuration of the higher level cognitive cone the energy of comments I see you're writing furiously no um drawing figures doodling seriously um okay so bishan goes on to say talk about a salience trap he says while salience is critical to an active inference it can nonetheless potentially increase free energy to empower abductive sense making so that weak signals can serve as an infrari information Rich

enhancement to predictions processing listening to it wants to emerge organizational design is challenged to neutralize the tendency of experienced managers and board members to overconfidently superimpose the potential fallacy of their historic and self-confidence-powered pattern recognition since the human brains task positive Network and default mode Network spheres are antagonistically related the only way to embrace antagonism as Duality is by linking in circular causality top-down predictions and bottom-up stimuli um and so here it's it I think that um Bijan is really uh maybe kind of criticizing or or like the maybe not criticizing maybe uh I don't know maybe criticizing's the right word like how experience leads you to prioritize like leads you to um Place importance on things that and maybe perhaps miss things that are important in an Adaptive organization I don't know do you guys have comments here we're drawn to the salience not to the meaningful or adoptive necessarily we hope that our salience field can be aligned with what's adaptive and meaningful but I think we found out that that is not always the case and it's funny like I think about just in scientific discovery right like how we're so attuned we're looking for the signal that we're looking for um it takes like a very perceptive and Adept scientist to not fall into that salience trap and perhaps notice um what's going on in the periphery right like even though this is the center this is what we're focusing on there's a lot of peripheral things happening that that might actually be important or as important or maybe more important than than the central thing that we're focused on um yeah I mean I guess that kind of the other thing I say is that this is just like another it's one of those things where I agree with this in Prince and like theoretically but there's so much stress going to like many of these decisions like for example if you have to do layouts for example like it's very time sensitive and the stakes are really high and so you know going off your previous experience can be really helpful in there because you have to just make it really quick decision and it's often like going to be heuristic space uh which may be not necessarily optimal but you really are often constrained by like your time and resources and like really processing all of your bottom up seeing the signals like even with the best intentions is really hard to do um and so again it's not to say like you shouldn't try to do it or it can't be done but it's just like it's another one of these things we're like hey this is a really big ass and I just want to call out like this is really really really really really hard to do especially when you're under under the gun yeah let's see this is in the ideal world this is how it would work right um okay do you want to read this claim first Tyler sure um all right so indeed this real-time switching vacillation process cannot be relegated to a single person if you need a CEO or executive chairman or upper echelon as a team but must be the result of an

unsupervised and FTP determined learning process that relates to top-down predictions and bottom of data input circular causality to vacate resource reconfiguration accordingly consequently the board's potentially unique dmn Dimension cannot be ignored but must be integrated into the process at the very top the board of directors has given us cognitive diversity and composition and cognitive distance firms done at tpn exploitation Logics emerges as a monitoring surveyor of the system that is centered around managing the dynamic interplay of dual forces exploitation exploration and financial and strategic controls um yeah yeah something I thought that was interesting here is that like you know while he's making this claim and which he's been through the whole book that like pay the boards in this really neat position to be just kind of like an integrated unit it kind of comes back to what we were saying earlier of like okay who was actually doing this do you want to be management and so in one sense I'm like okay it makes sense that the board would be doing this because you can hold them accountable to holding these different ideas in their head at the same time uh managing Duality but it seemed I would think that you would need that Duality management to be going down the hierarchy not just at the border as well and at each level the hierarchy is managing duality in some sense not from so I very much want to ask you on that when we uh got on a call with him blue day one thing to add here no um but this is one sentence that I pulled out randomly um from the chapter that just like struck me and like he just like threw it in here and didn't unpack it through anything the firm is the generative social field and emerges in language and I thought that this this was like really profound so um I don't know I just put it in here because like just like wow that's very deep um I don't know I don't know if you guys have comments or this is something like I would ask John to expand upon you could ask a hundred people with different backgrounds what this invokes for them and see where that goes that sounds like a fun project it was just like a like a sentence that seemed very like yeah deep and like just casually thrown in there yeah so I'm gonna get this tattooed on my chest right put it on your next business card that'd be awesome that'd be an awesome like statement for a business card right um that'd be great so what is it you say you do I don't know I guess that's true this is the business philosophy so do you guys have any final like comments questions Tyler did you tell us all the things you wanted to tell us no that's it for me yeah we're moving along through the structure I'm looking forward to more conversations and also some digestions and Reflections making some schemes to visualize at a high level what we've been coming away from because this is like our eighth session and I think we're starting to see his patterns and style of laying out what he wants to convey and integrate yeah I definitely think

we're getting like the repeat like repeat the message and so so it's starting to drive some of it home for me at least so coming up to the end of the second part of the book next week and I'll be looking forward to that and um to hopefully discuss soon with the author at his convenience that that part all right well thank you blue and Tyler farewell bye bye next week same time